Resources

for

Probability & Statistics

BITS PILANI DIGITAL

Reference Materials for Probability and Statistics

1. OpenStax Introductory Statistics (2nd Edition)

OpenStax Introductory Statistics

Publisher: OpenStax (Rice University)
License: Creative Commons Attribution License
Level: Undergraduate Introduction
Prerequisites: Intermediate Algebra (Class 12 Math)

How to Access:

1. Visit: `https://openstax.org/details/books/introductory-statistics-2e`
2. Click on "**View Online**"
3. Then click "**Go to your book**"
4. *Alternative:* You can also download the book directly from the same page

2. Introduction to Probability for Data Science

Introduction to Probability for Data Science

Publisher: University of Michigan Publishing
Year: 2021
Authors: University of Michigan Faculty
Specialization: Data Science Applications

How to Access:

1. Visit: `https://probability4datascience.com/`
2. On the left panel, you will find different chapters
3. Click on any chapter to read the contents
4. Navigate through the material using the chapter links

3. Introduction to Probability, Statistics, and Random Processes

Probability, Statistics & Random Processes

Author: Hossein Pishro-Nik
Institution: University of Massachusetts Amherst
Type: Open access peer-reviewed textbook
License: Creative Commons Licensed

How to Access:

1. Visit: `https://www.probabilitycourse.com/`
2. On the left panel, click on different chapters
3. Click on the subsections within each chapter
4. Use **"Next"** and **"Previous"** buttons to navigate through the contents

4. An Introduction to Statistical Learning

Statistical Learning with R and Python

Authors: James, Witten, Hastie, Tibshirani
Publisher: Springer (Free with author permission)
Editions: Available in both R and Python versions
Focus: Machine Learning and Statistical Methods

How to Access:

1. Visit: `https://www.statlearning.com/`
2. Choose between:
   - **R Version:** Click **"Download ISL with R"**
   - **Python Version:** Click **"Download ISL with Python"**
3. Free PDF downloads available directly from the authors' official website

5. Probability and Statistics for Data Science (NYU)

Probability and Statistics for Data Science

Author: Carlos Fernandez-Granda
Institution: New York University (NYU)
Department: Center for Data Science
Publisher: Cambridge University Press (2025)
Status: Free preprint made available by author

How to Access:

1. Visit the official book website: `https://www.ps4ds.net/`
2. Click on **"A free preprint"** link
3. You will be directed to the GitHub repository
4. Download the PDF: `https://github.com/cfgranda/ps4ds/blob/main/ps4ds_preprint.pdf`
5. *Additional Resources on the website:*
    - 102 Python notebooks using 23 real-world datasets
    - 115 videos with slides explaining key concepts
    - Solutions to 200 exercises

6. Khan Academy Statistics and Probability

Khan Academy Statistics and Probability

Platform: Khan Academy
Format: Video lessons with practice exercises
Level: Beginner to intermediate
Access: Completely free with user account

How to Access:

1. Visit: `https://www.khanacademy.org/math/statistics-probability`
2. Browse through different topics and lessons
3. Watch video explanations and complete practice exercises
4. Track your progress through the interactive platform

Topics Covered Across These Resources

Core Topics for Data Science & AI

- **Descriptive Statistics:** Mean, median, mode, variance, standard deviation
- **Probability Theory:** Basic probability, conditional probability, Bayes' theorem
- **Probability Distributions:** Normal, binomial, uniform, exponential distributions
- **Random Variables:** Discrete and continuous random variables
- **Statistical Inference:** Confidence intervals, hypothesis testing
- **Regression Analysis:** Linear and logistic regression, model selection
- **Machine Learning Methods:** Classification, clustering, tree-based methods
- **Resampling Methods:** Cross-validation, bootstrap methods
- **Advanced Topics:** Support vector machines, neural networks, deep learning
- **Data Visualization:** Histograms, scatter plots, statistical graphics

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